NSRIT

Assignment Question Paper

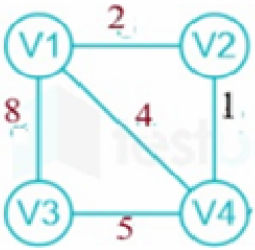
Degree	B. Tech.	Branch	CSD	Assignment No.	III	Academic Year	2026-27
Course Code	23CS601			Max. Marks	30	Semester	V
Course Name	Computer Networks						

Answer any three of the following questions
Each question carries equal Marks

SET – 1

Q. No.	Questions (1 through 3)	Marks	Learning Outcome (s)	DoK
1	a) Elucidate link state routing algorithm.with an example **[5M]** b) Give the general principles of congestion control in detail **5M**	10M	23CS601.3	L1 – L3
2	Discuss in detail fragmentation in network layer **[5M]** Explain the purpose of tunneling in Internet Layer **[5M]**	10M	23CS601.3	L1 – L3
3	(a) Given a network address of 192.168.100.0 and a subnet mask of 255.255.255.192. **i)** How many subnets are created? **ii)** How many hosts are there per subnet? **iii)** What is the broadcast IP and network ranges of each subnet? **[5M]** (b) Explain classful addressing and subnet mask. Why does class D have no network or host id **[5M]**	10M	23CS601.3	L1 – L3

SET – 2

Q. No.	Questions (4 through 6)	Marks	Learning Outcome (s)	DoK
4	Apply distance vector algorithm for the graph:shown below:  How is distance vector algorithm different from link state algorithm?	10M	23CS601.3	L1 – L3
5	Briefly explain the leaky bucket and token bucket algorithms with neat sketches	10M	23CS601.3	L1 – L3
6	a) Draw the IPv4 header format and explain the significance of all the fields **[5M]** b) Mention any main 4 differences between IPV4 & IPV6 **[5M]**	10M	23CS601.3	L1 – L3

SET – 3

Q. No.	Questions (7 through 9)	Marks	Learning Outcome (s)	DoK
7	Give an example to mention the purpose of subnet mask in IPv4 address? **[5M]**	10M	23CS601.3	L1 – L3

	What is classful addressing? Illustrate subnetting in networks with classful and classless addressing with examples. **[5M]**			
8	What is congestion? Explain the various approaches to congestion control. What is the role of choke packet and load shedding in managing congestion?	10M	23CS601.3	L1 – L3
9	a) Illustrate Hierarchical routing algorithm with a example **[5M]** b) Write in detail about OSPF and BGP algorithms **[5M]**	[10M]	23CS601.3	L1 – L3